

Report generated on

PVGIS-5.3 geo-temporal irradiation database

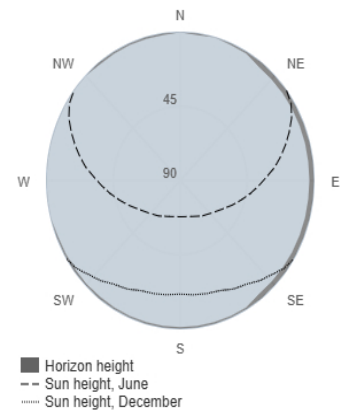
Provided inputs

Latitude/Longitude: 45.437,12.335
Horizon: Calculated
Database used: PVGIS-SARAH3
Start year: 2005
End year: 2023

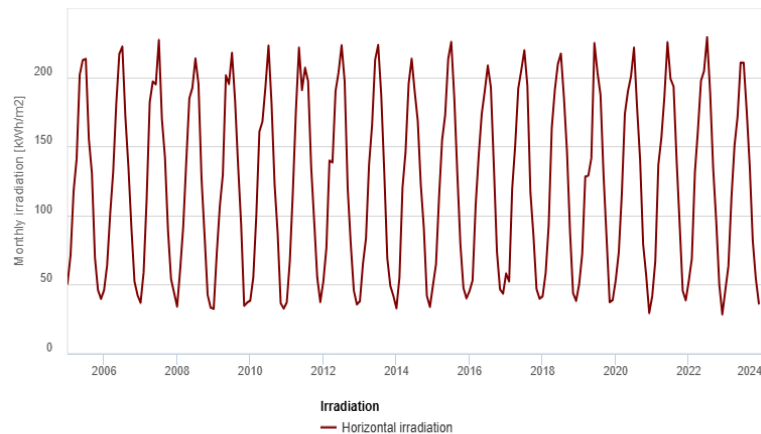
Variables included in this report:

Global horizontal irradiation: Yes
Direct Normal Irradiation: No
Global irradiation optimum angle: No
Global irradiation at angle °: No
Diffuse/global ratio: Yes
Average temperature: Yes

Outline of horizon at chosen location:



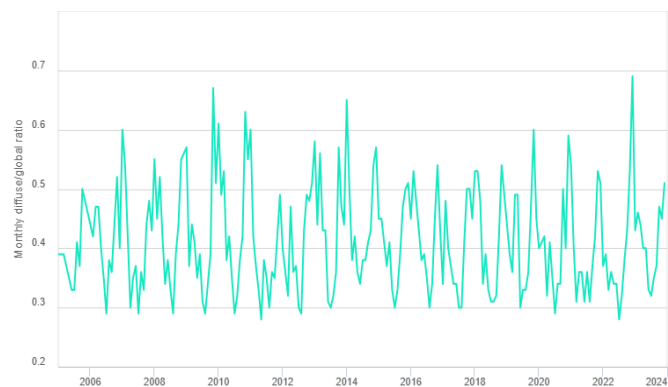
Monthly solar irradiation estimates



Global horizontal irradiation

Month	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
January	50.8	45.96	36.89	34.16	32.52	38.51	37.13	52.31	38.01	32.79	49.51	45.33	57.95	41.44	50.32	53.24	41.84	52.78	46.15
February	71.37	64.01	58.92	60.99	72.42	55.2	66.99	76.35	64.96	55.37	64.89	52.98	52.29	58.26	72.15	73.26	66.83	68.9	63.39
March	117.66	100.89	112.4	92.16	106.54	100.41	117.41	139.65	83.54	120.11	113.47	107	119.21	93.09	128.09	115.56	136.67	130.97	112.92
April	141.13	131.88	181.82	139.17	129.64	160.66	175.83	138.36	136.79	146.02	154.63	143.03	151	162.98	128.77	173.65	156.16	161.8	150.44
May	201.62	180.66	196.84	184.67	201.22	167.78	221.31	189.78	165.38	194.85	172.49	173.44	191.9	190.36	141.52	189.74	186.27	197.41	171.29
June	212.38	216.72	194.86	192.24	195.03	193.14	190.7	203.62	212.58	213.36	213.23	190.06	205.04	209.32	224.55	200.47	225.26	204.45	210.53
July	213.2	222.1	226.86	213.51	217.5	222.77	206.7	223.1	223.21	189.64	225.4	208.35	219.42	217	203.55	221.39	198.74	228.83	210.42
August	155.4	171.2	169.8	195.2	183.49	180.79	197.09	196.67	187.26	168.54	184.9	192.61	193.84	185.28	187.29	178.41	193.35	189.13	175.79
September	130.67	134.97	142.06	126.79	136.42	122.49	135.37	119.12	131.3	122.35	129.77	134.76	116.55	146.28	129.24	140.37	139.95	133.95	135.56
October	69.58	90.35	89.16	85.57	93.67	87.25	95.58	79.36	68.91	90.94	78.92	73.52	85.25	88.07	83.56	79.15	93.47	95.89	81.43
November	45.98	52.06	53.95	42.68	34.69	36.5	55.01	45.75	49	41.74	47.52	46.52	46.76	43.72	37.29	56.73	45.43	50.06	54.2
December	39.66	42.35	43.96	33.36	37	32.67	37.42	35.64	41.78	33.89	40.04	43.43	39.81	38.24	38.91	29.44	38.68	28.43	36.32

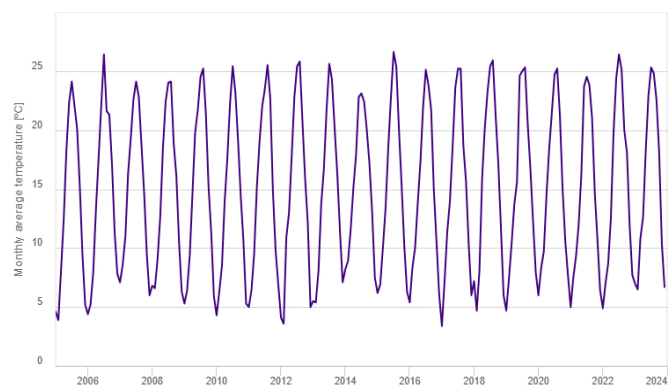
Monthly average diffuse to global ratio



Diffuse/global ratio

Month	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
January	0.39	0.44	0.6	0.55	0.57	0.61	0.6	0.4	0.58	0.65	0.45	0.45	0.34	0.53	0.44	0.4	0.54	0.37	0.43
February	0.39	0.42	0.54	0.45	0.37	0.49	0.42	0.36	0.44	0.5	0.45	0.53	0.48	0.53	0.39	0.41	0.4	0.39	0.46
March	0.39	0.47	0.42	0.52	0.44	0.53	0.37	0.32	0.56	0.38	0.41	0.48	0.4	0.48	0.36	0.42	0.31	0.33	0.44
April	0.37	0.47	0.3	0.42	0.41	0.38	0.33	0.47	0.43	0.42	0.37	0.43	0.37	0.34	0.49	0.32	0.36	0.36	0.4
May	0.35	0.4	0.35	0.34	0.35	0.42	0.28	0.36	0.43	0.36	0.41	0.38	0.34	0.39	0.49	0.41	0.36	0.34	0.4
June	0.33	0.35	0.37	0.38	0.39	0.35	0.38	0.37	0.31	0.34	0.33	0.39	0.34	0.33	0.3	0.35	0.31	0.34	0.33
July	0.33	0.29	0.29	0.33	0.31	0.29	0.35	0.3	0.3	0.38	0.3	0.35	0.3	0.31	0.33	0.29	0.36	0.28	0.32
August	0.41	0.38	0.36	0.29	0.29	0.32	0.3	0.29	0.32	0.38	0.33	0.3	0.3	0.31	0.33	0.34	0.31	0.32	0.35
September	0.37	0.36	0.33	0.39	0.34	0.38	0.36	0.43	0.36	0.41	0.39	0.34	0.41	0.32	0.36	0.34	0.37	0.38	0.37
October	0.5	0.44	0.44	0.44	0.39	0.42	0.35	0.49	0.57	0.43	0.47	0.45	0.5	0.42	0.47	0.5	0.42	0.43	0.47
November	0.48	0.52	0.48	0.55	0.67	0.63	0.42	0.48	0.47	0.54	0.5	0.54	0.5	0.54	0.6	0.4	0.53	0.53	0.45
December	0.46	0.4	0.43	0.56	0.51	0.55	0.49	0.51	0.44	0.57	0.51	0.43	0.45	0.49	0.45	0.59	0.51	0.69	0.51

Monthly average temperature



Monthly average temperature

Month	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
January	4.6	4.4	7.1	6.8	5.3	4.3	5	4.1	5.5	8.2	6.2	5.4	3.4	7.2	4.7	6	5	4.9	7
February	3.9	5.2	8.6	6.6	6.4	6.3	6.4	3.6	5.4	8.9	6.9	8.3	7.2	4.7	7.4	8.3	7.5	7	6.5
March	8.1	7.9	11	9.1	9.4	8.5	9.5	10.9	8.1	11.6	10	10	11.4	8.1	10.4	9.7	9.3	8.7	10.8
April	12.4	13.2	16.3	12.7	14.6	13.9	15.1	13	13.7	15.1	13.4	13.8	14	15.9	13.7	14.5	12	12.5	12.6
May	18.3	17.6	19.3	18.5	19.8	17.6	19	17.8	16.7	17.9	18.5	17.3	18.6	20.2	15.6	18.4	16.3	19.8	18.2
June	22.3	22.1	22.6	22.4	21.7	22.2	22	22.8	21.7	22.8	22.8	22	23.6	23.2	24.6	21.5	23.7	24.4	22.9
July	24.1	26.4	24.1	24	24.5	25.4	23.5	25.4	25.6	23.1	26.6	25.1	25.2	25.4	25	24.7	24.5	26.4	25.3
August	22.1	21.6	22.8	24.1	25.2	23.2	25.5	25.8	24.3	22.4	25.4	23.8	25.2	25.9	25.3	25.2	23.8	25.1	24.8
September	20.1	21.3	18.7	18.9	21.4	19.1	22.7	20.8	20.2	20	20	21.7	18.7	21.4	20.5	21.2	21	20	22.6
October	15.2	17.1	14.6	16.1	15.2	14.4	14.8	15.9	16.3	17.1	15.2	14.9	15.6	17.3	16.7	15	14.7	18.1	18.1
November	9.4	11.3	9.4	10.5	11.3	10.6	9.8	12.1	11.4	13	10.1	10.5	10.3	11.8	12.2	10.6	10.7	11.9	10.4
December	5.2	7.8	6	6.3	5.9	5.3	6.9	5	7.1	7.5	6.3	6	6	6	7.9	7.7	6.4	7.7	6.7